LAB WORKSHEET

IDS / IPS Lab

Detect, then block, attacks with Snort: IDS on a SPAN port vs IPS inline.

What you learn

- **IDS:** out of band on a SPAN/mirror port; it can only **alert**. **IPS:** inline; it can **drop**.
- Signature vs anomaly detection; true/false positives and negatives.
- Rule anatomy: `action proto src_ip src_port -> dst_ip dst_port (options)`

Key syntax

```
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ICMP Ping detected"; itype:8; sid:1000001;
rev:1;)
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (msg:"SSH brute force"; flags:S;
detection_filter:track by_src, count 5, seconds 60; sid:1000002; rev:1;)
drop tcp $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SQL injection"; content:"' OR '1'='1"; nocase;
http_uri; sid:1000003; rev:1;)
```

Use sid 1000001 and above for your own rules, and end every option with `;`. In IDS mode a `drop` rule only alerts.

Commands

```
ip a
sudo nano /etc/snort/snort.conf          (ipvar HOME_NET 192.168.1.0/24)
sudo nano /etc/snort/rules/local.rules
sudo snort -T -c /etc/snort/snort.conf
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0                    (IDS)
sudo snort -Q --daq afpacket -i eth0:eth1 -A console -q -c /etc/snort/snort.conf (IPS)
cat /var/log/snort/alert                 Ctrl+C stops Snort
```

nano: **^O** save, **^X** exit, **^K**/**^U** cut/paste, **F6** search.

Guided task

Check the interfaces → fix HOME_NET → validate → ICMP rule → run as an IDS → ping sweep → SSH brute-force rule → SQL-injection rule → switch to IPS → change the rule to drop and run inline → the attack is blocked and normal browsing raises no alert.

Independent task

Moorland Outdoor Supplies (fictional): detect the port scan, block Telnet, alert on the bad domain, block directory traversal, detect the test-malware marker, catch SSH brute force — with no false positives.